

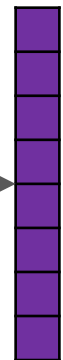
Vision Transformer (ViT)



What is in the image?



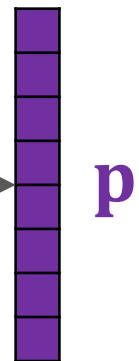
Neural
Network



p

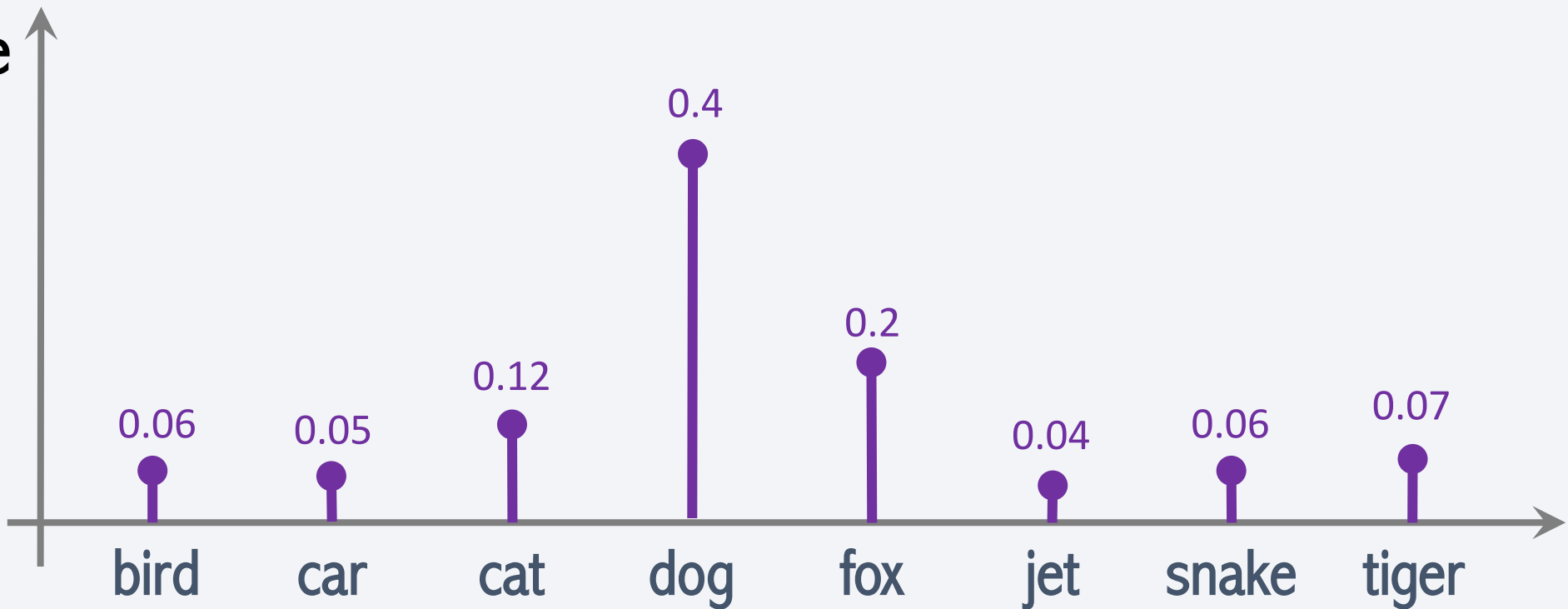


Neural
Network



p

Confidence



Classes

Image Classification

- CNNs, e.g., ResNet, were the best solutions to image classification.
- Vision Transformer (ViT) [1] beats CNNs (by a small margin), if the dataset for pretraining is sufficiently large (at least 100 million images).
- ViT is based on Transformer (for NLP) [2].

Reference

1. Dosovitskiy et al. [An image is worth 16×16 words: transformers for image recognition at scale](#). In *ICLR*, 2021.
2. Vaswani et al. [Attention Is All You Need](#). In *NIPS*, 2017.

Split Image into Patches

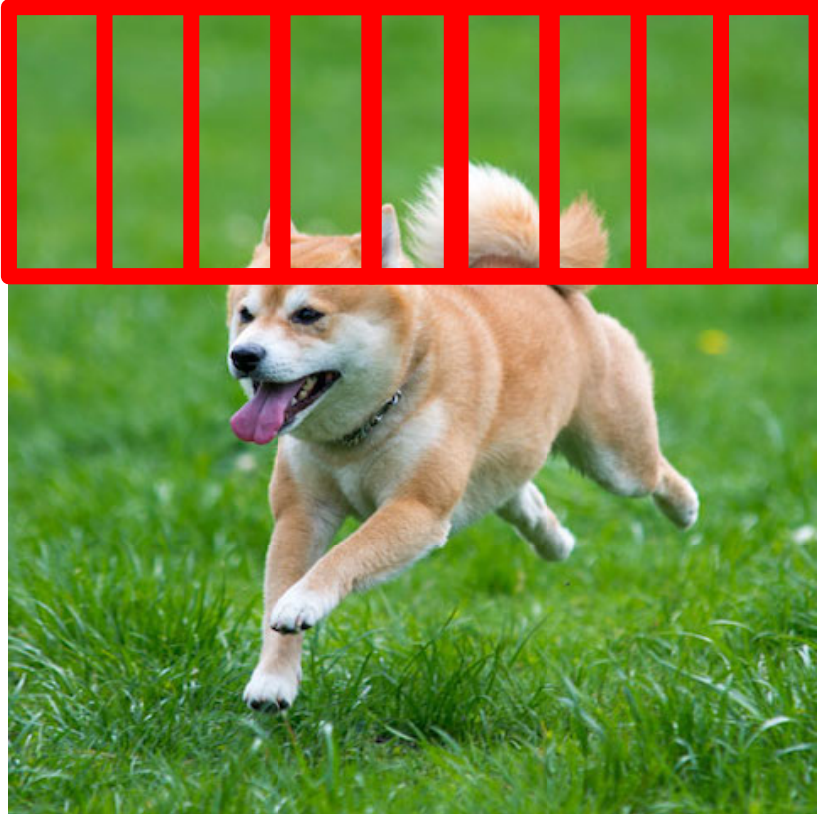


Split Image into Patches



- Here, the patches do not overlap.

Split Image into Patches

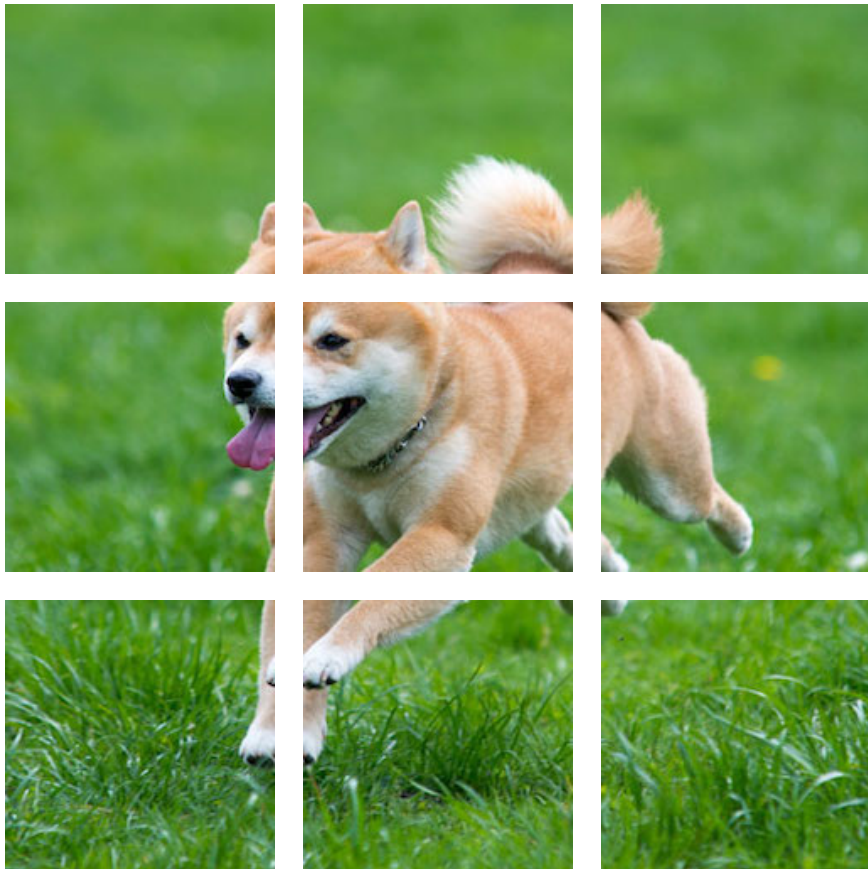


- Here, the patches do not overlap.
- The patches can overlap.
- User specifies:
 - **patch size**, e.g., 16×16 ;
 - **stride**, e.g., 16×16 .

Vectorization

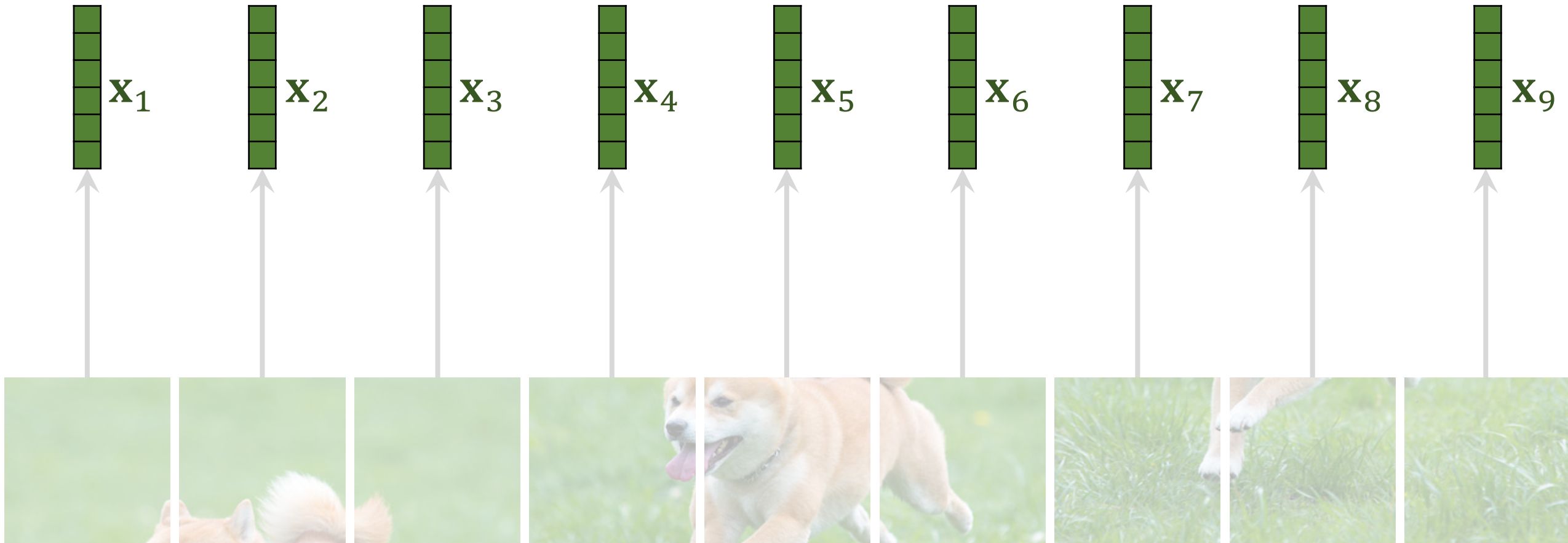


Vectorization



Vectorization

If the patches are $d_1 \times d_2 \times d_3$ tensors, then the vectors are $d_1 d_2 d_3 \times 1$.





$\mathbf{x}_1$



$\mathbf{x}_2$

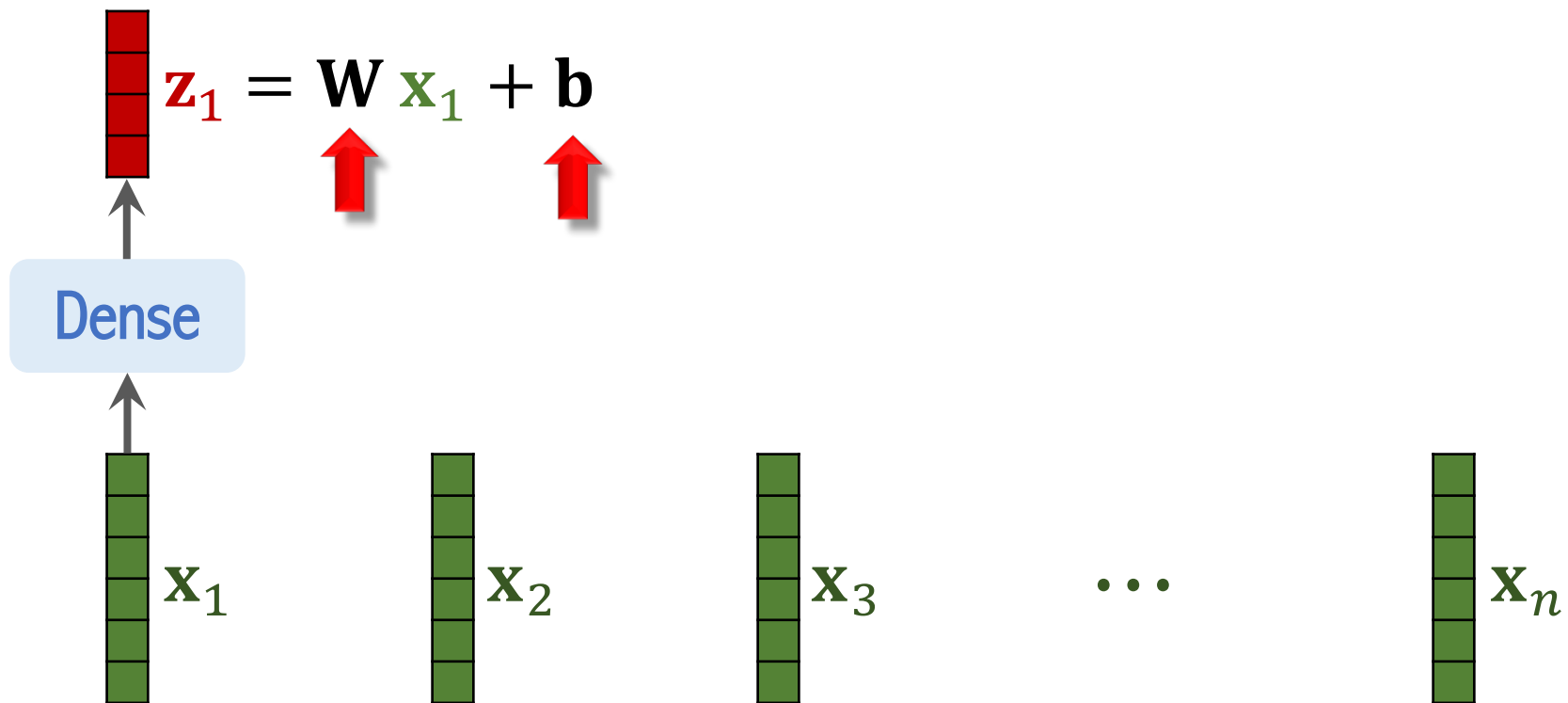


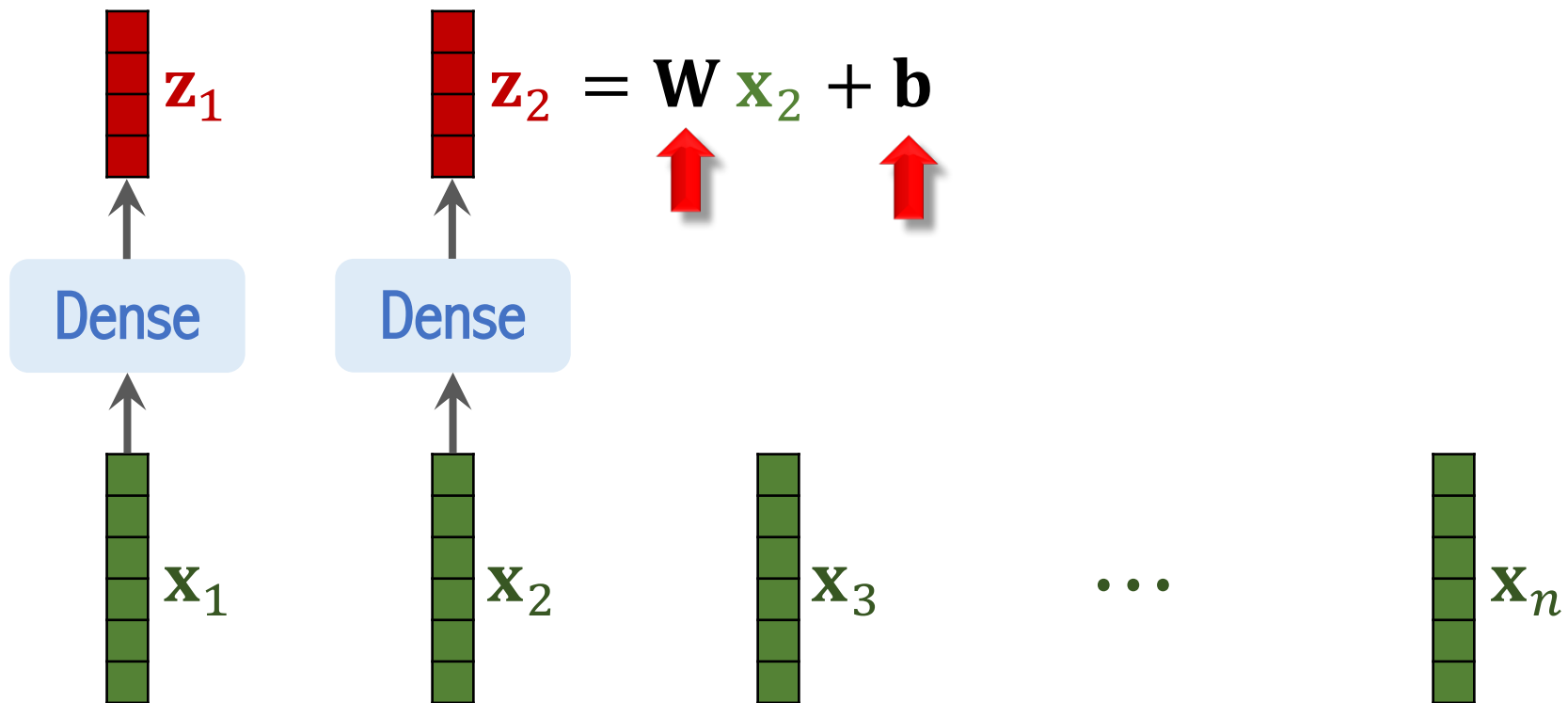
$\mathbf{x}_3$

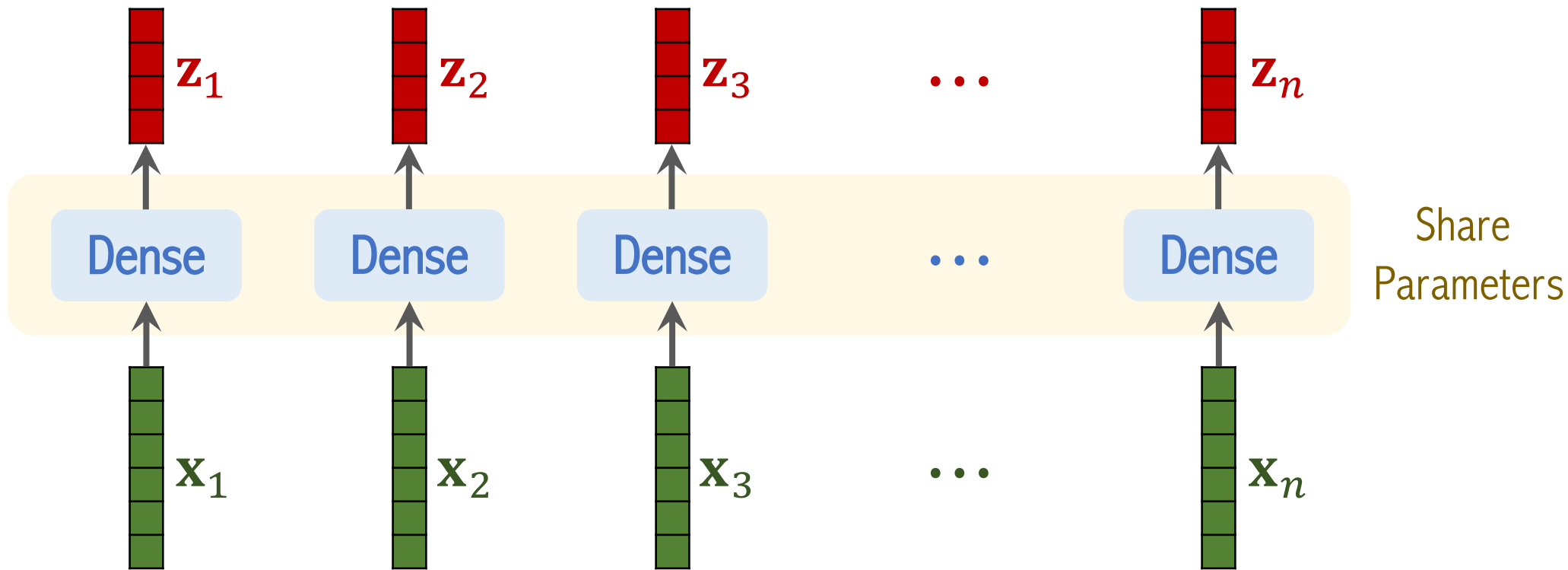
$\dots$



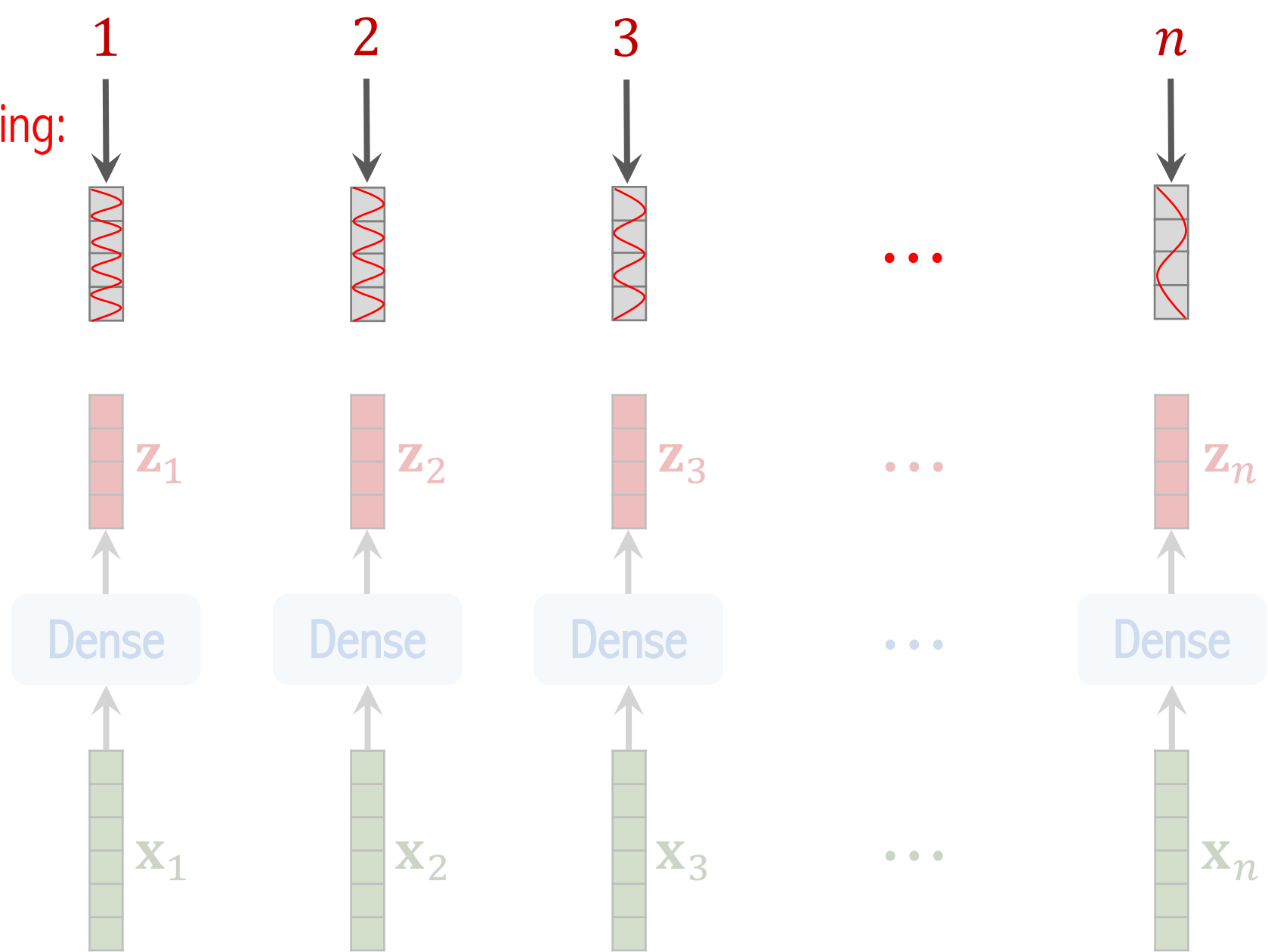
$\mathbf{x}_n$



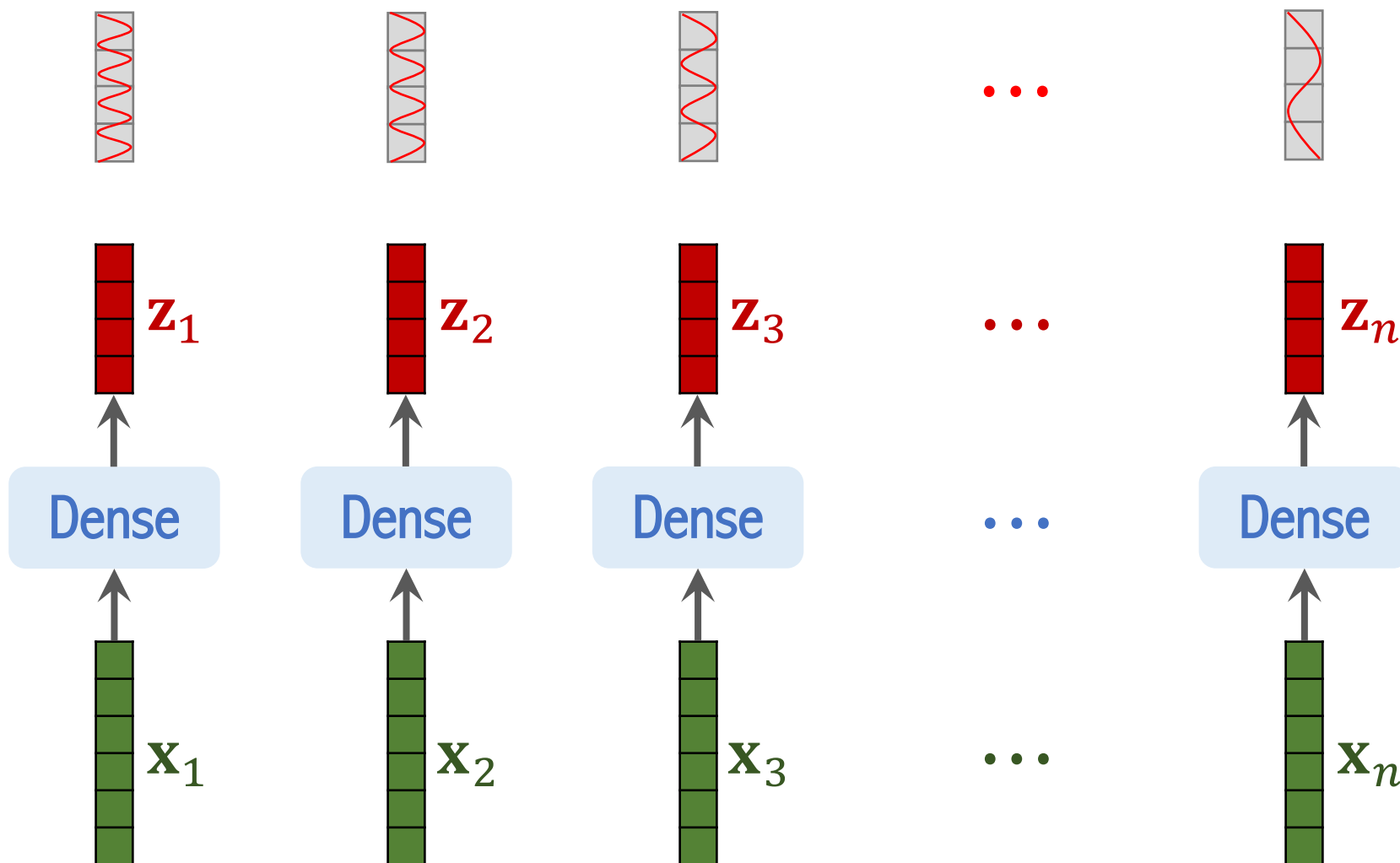




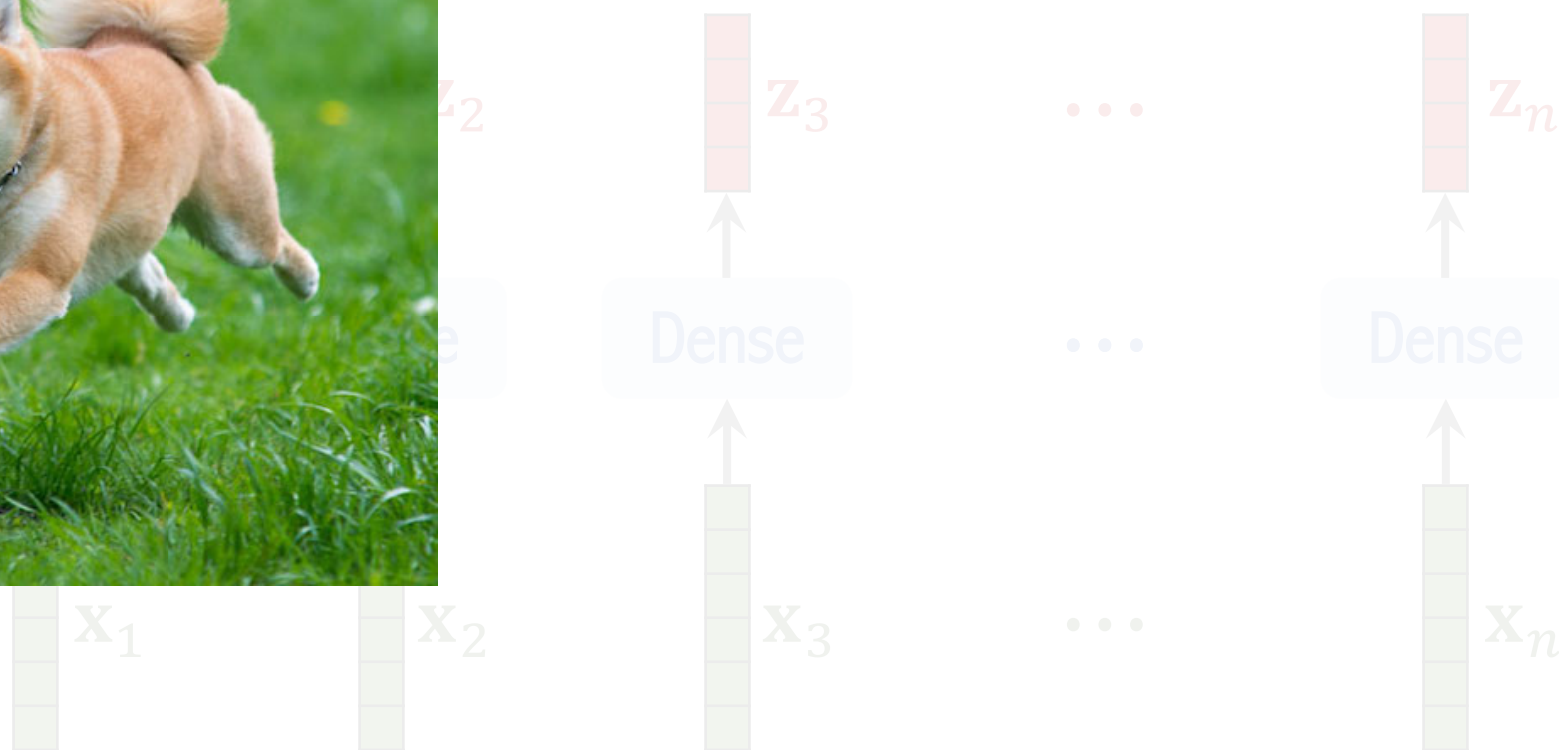
Positional Encoding:



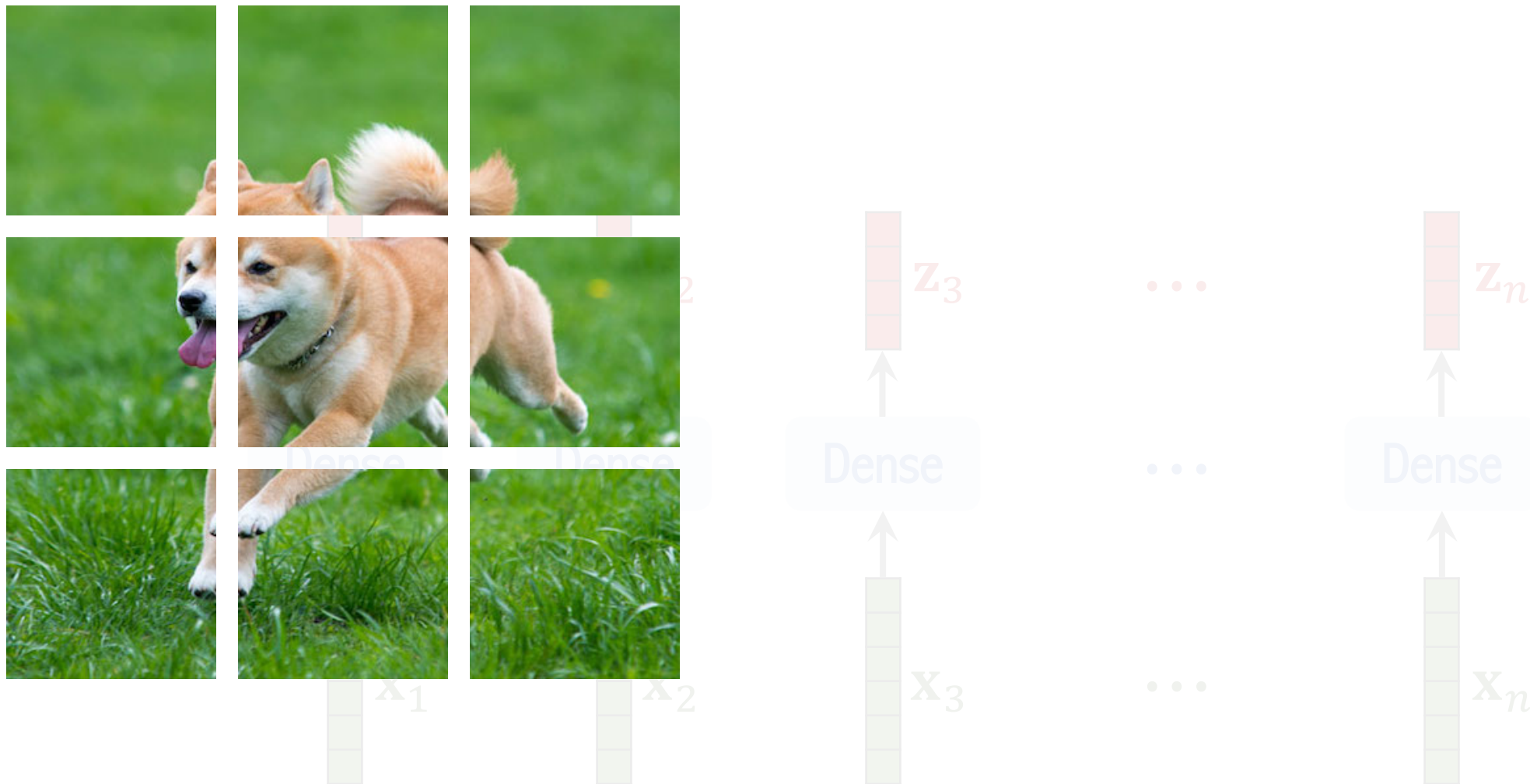
Add positional encoding vectors to $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_n$.



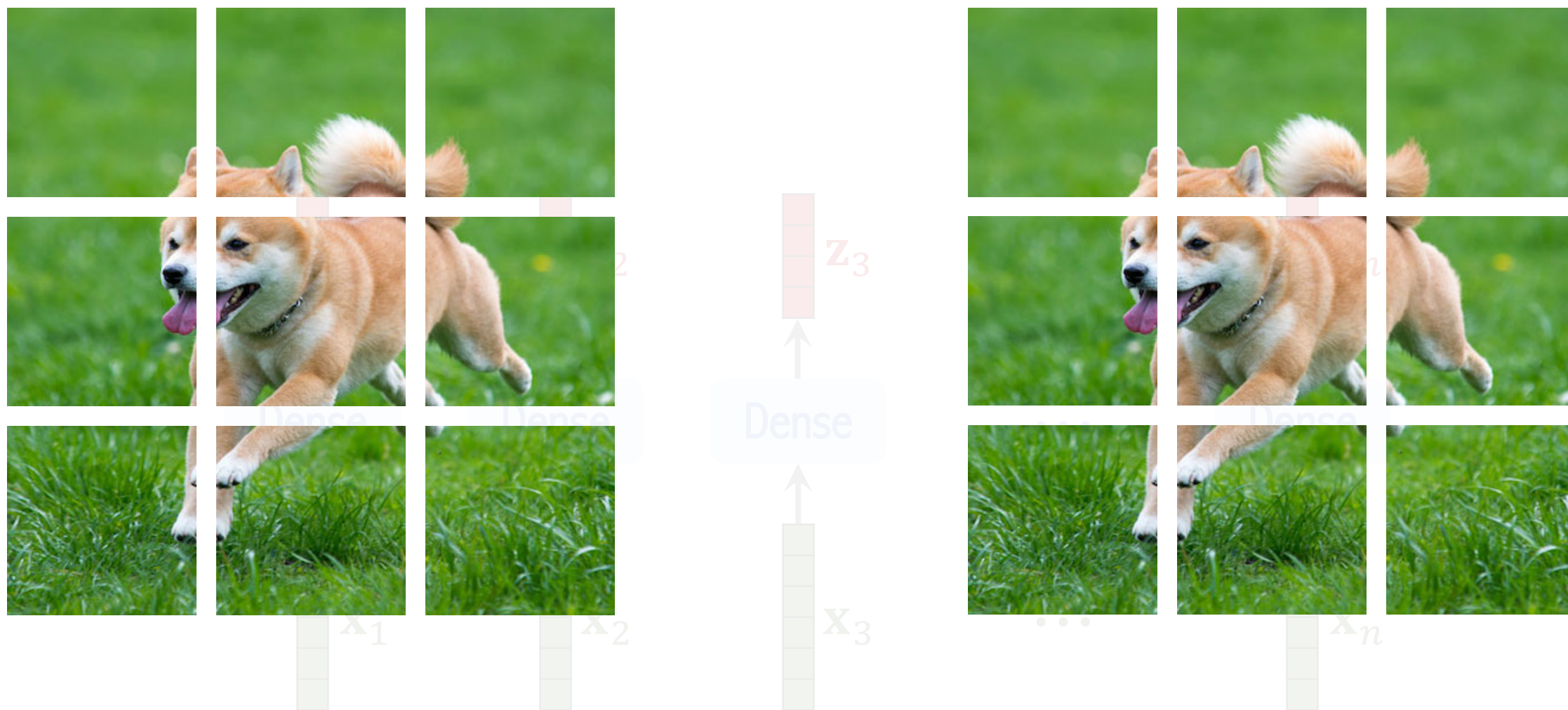
Add positional encoding vectors to $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_n$. (Why?)



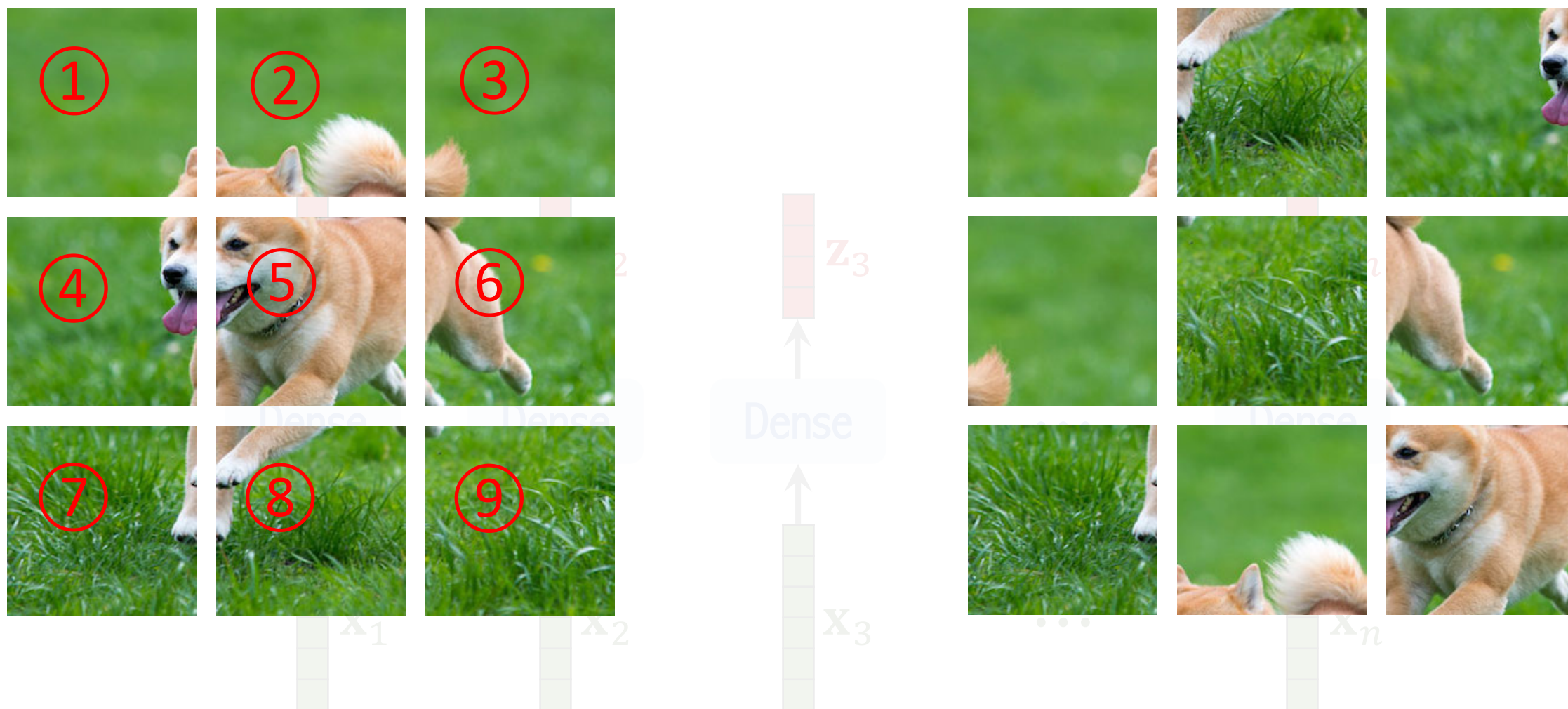
Add positional encoding vectors to $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_n$. (Why?)

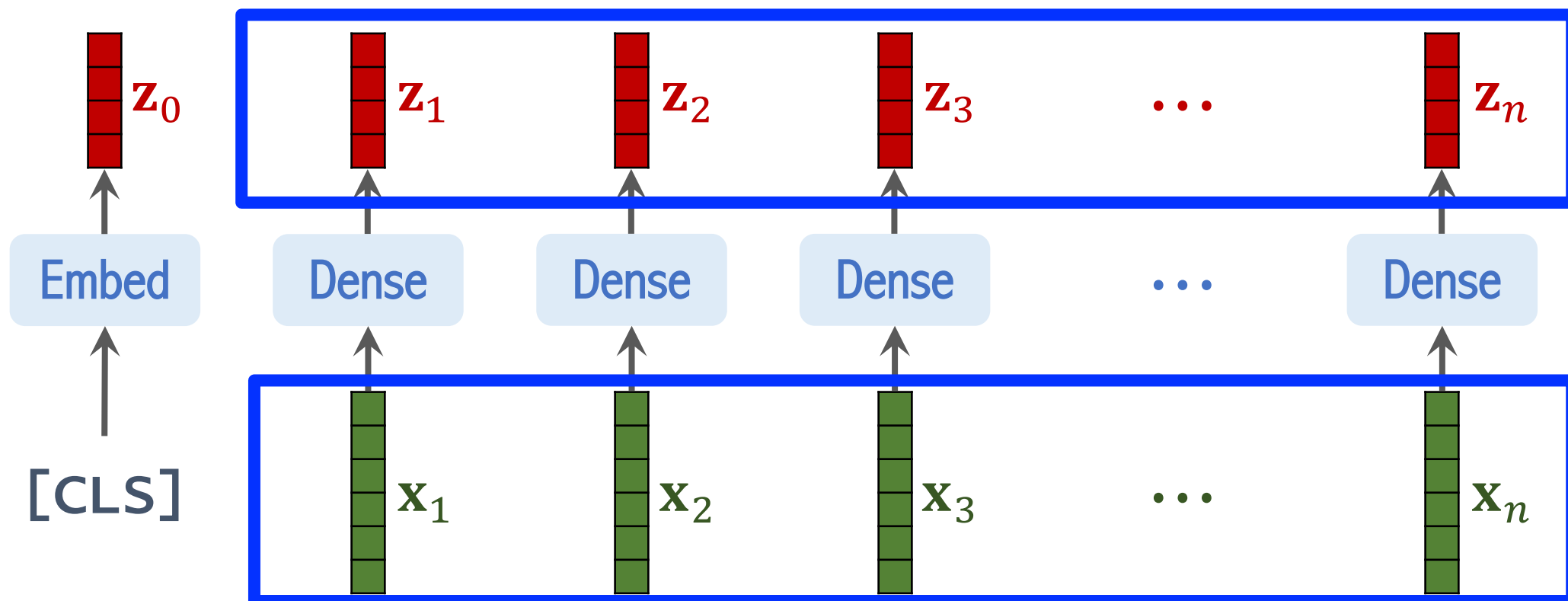


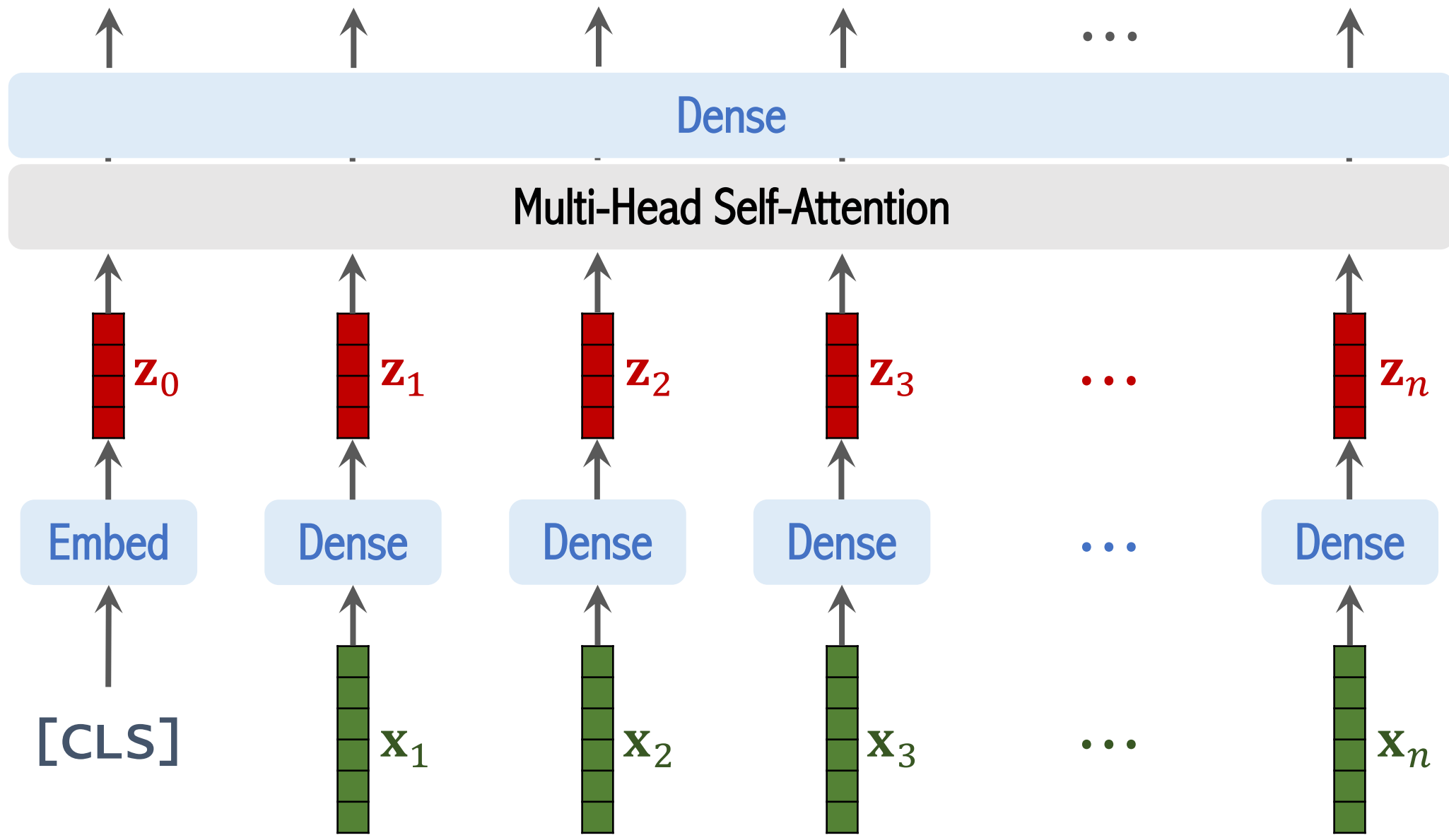
Add positional encoding vectors to $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_n$. (Why?)

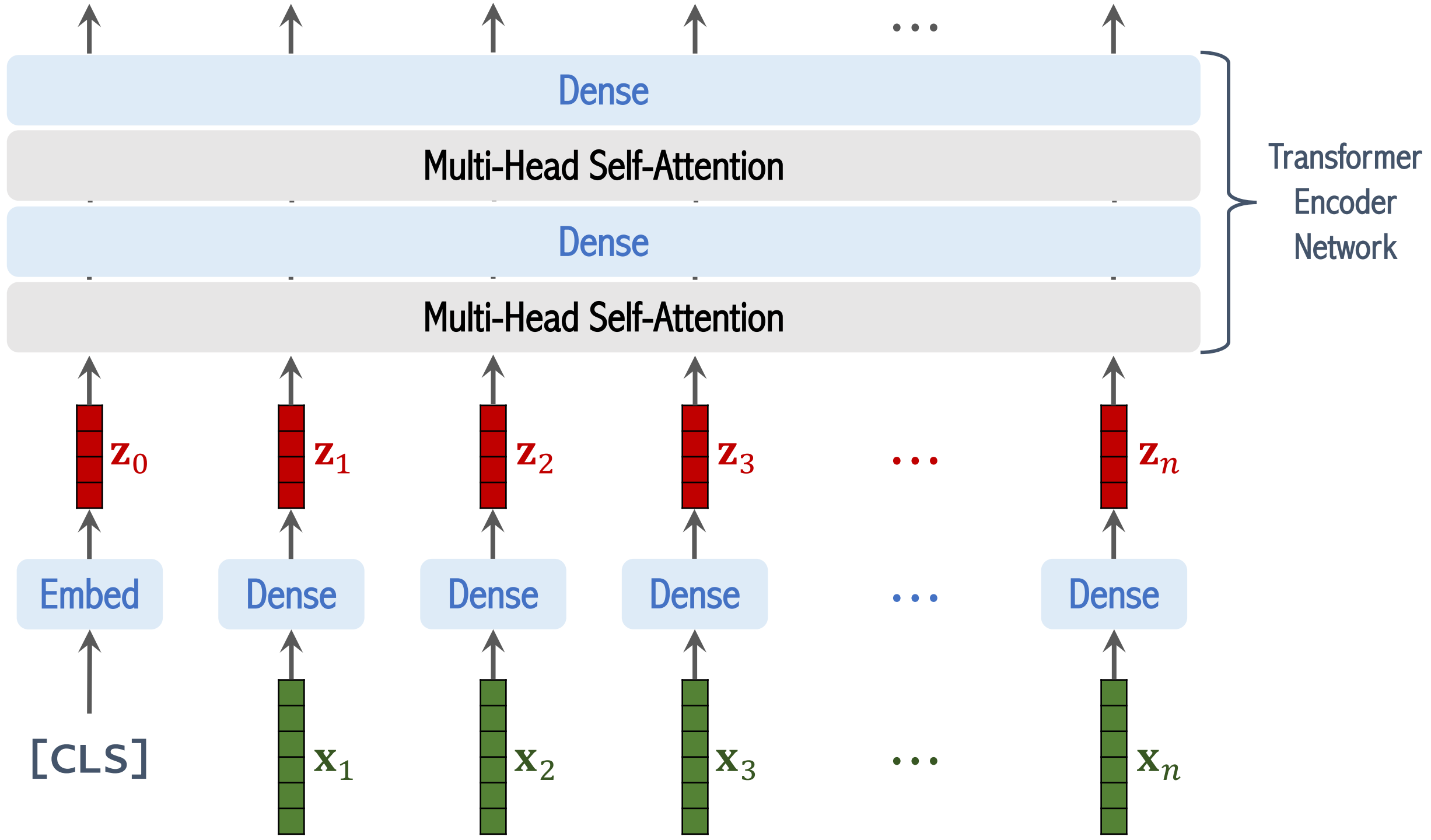


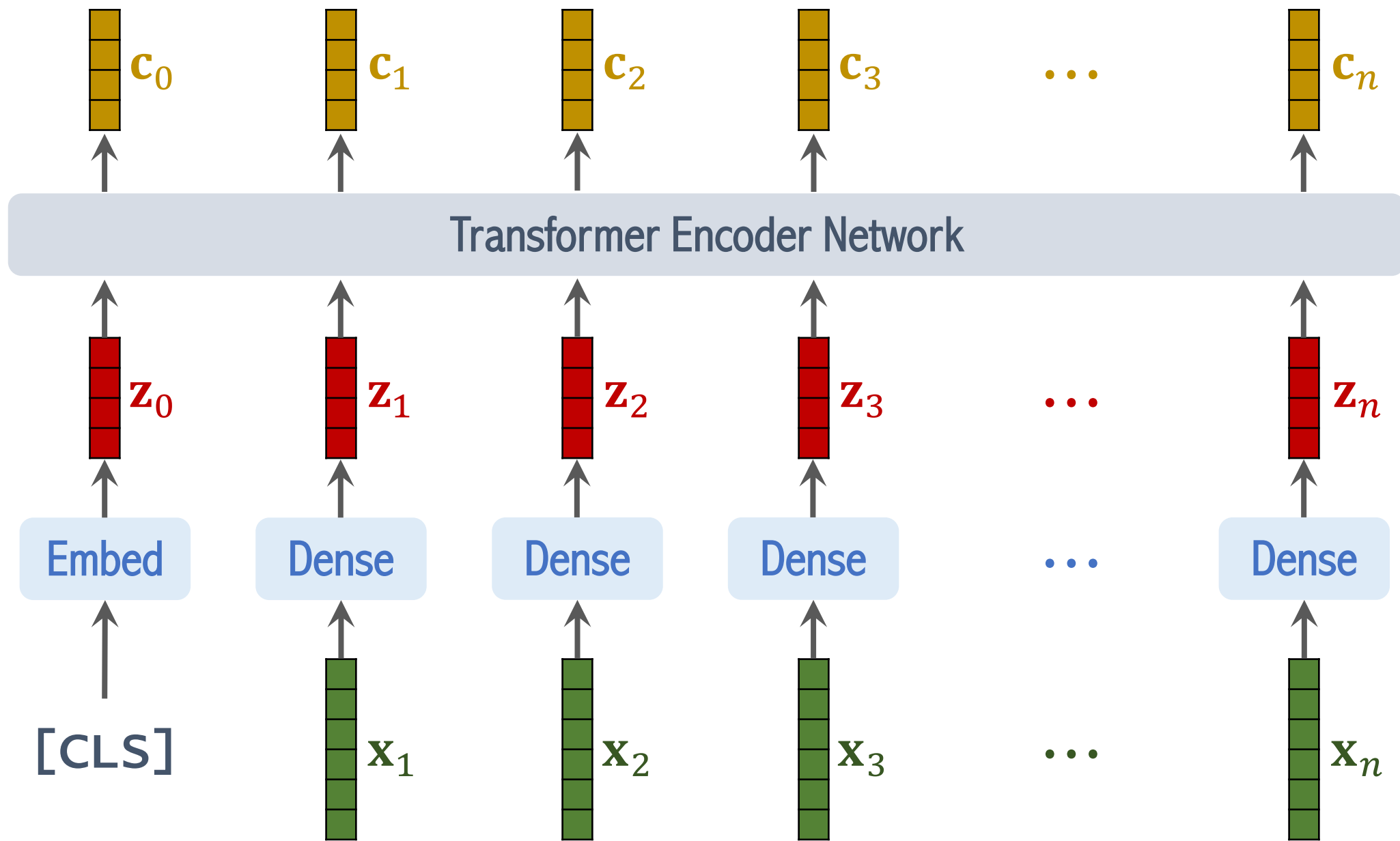
Add positional encoding vectors to $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_n$. (Why?)

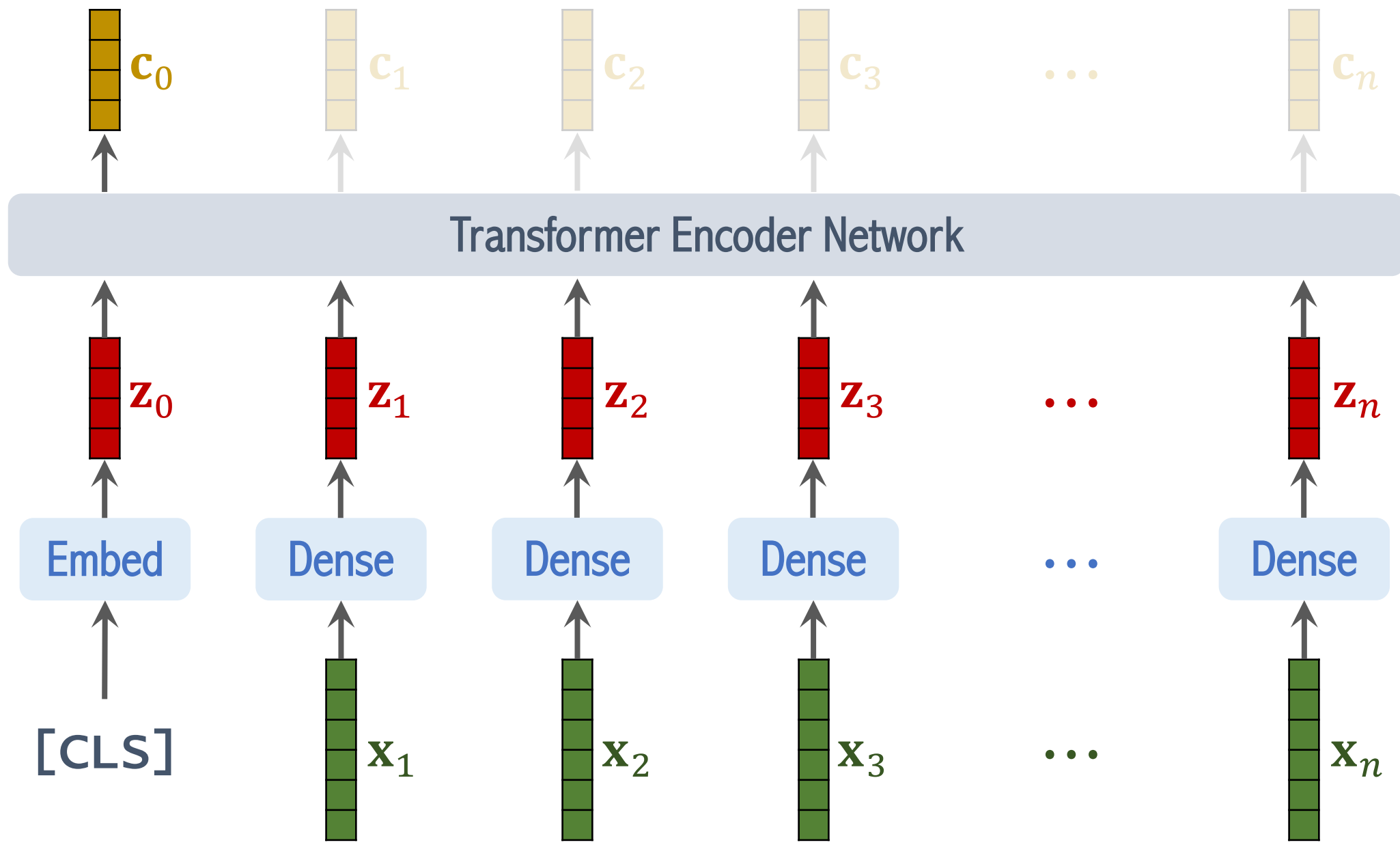


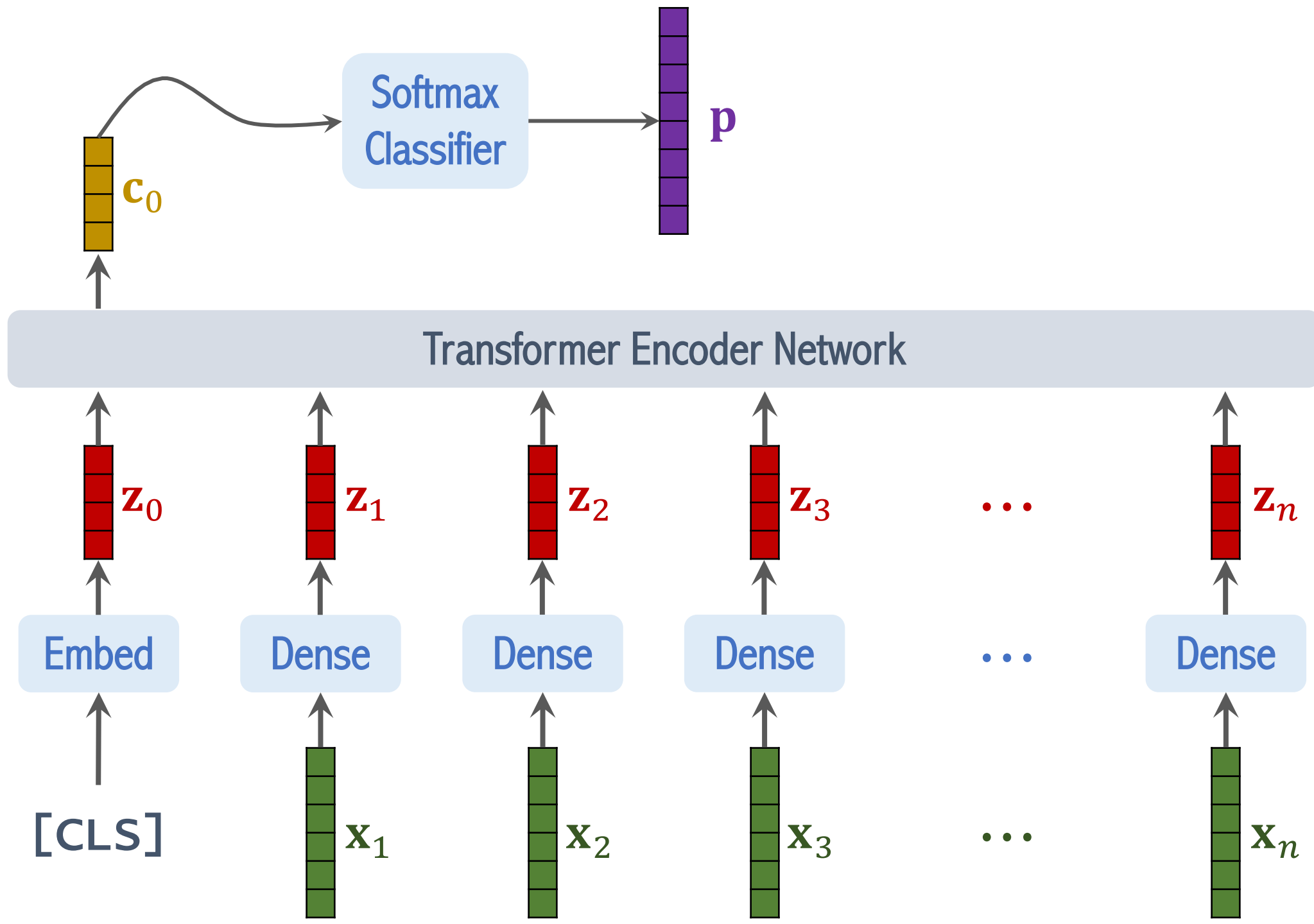


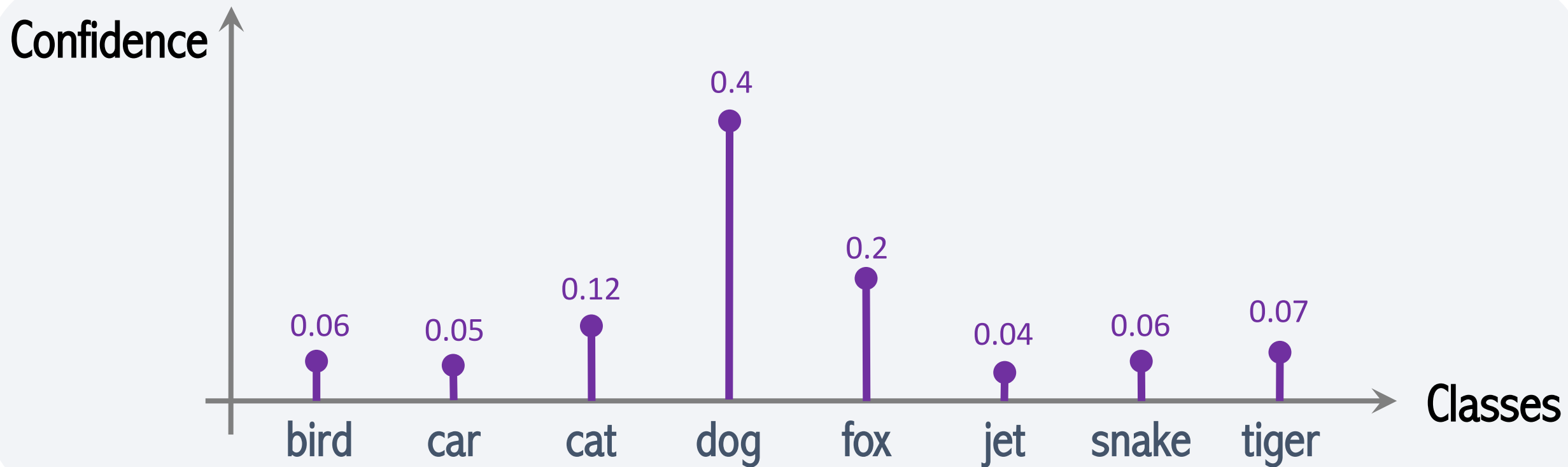
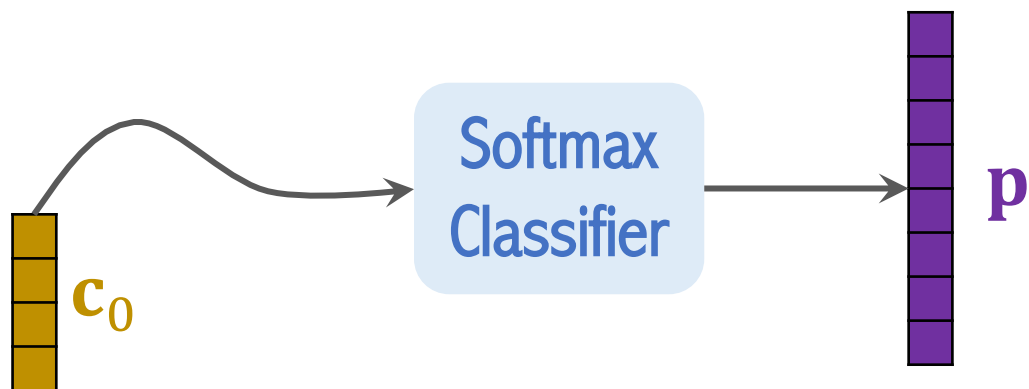




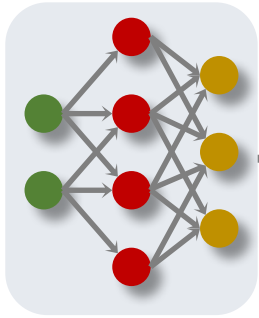




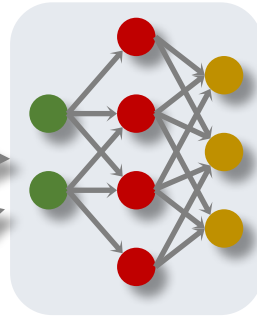




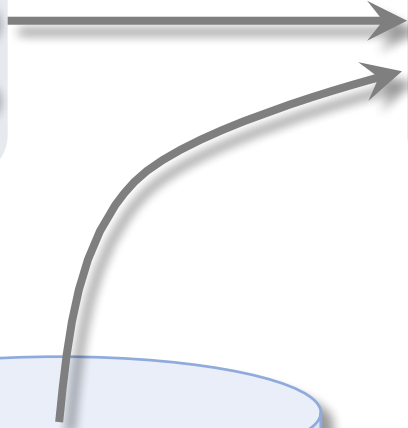
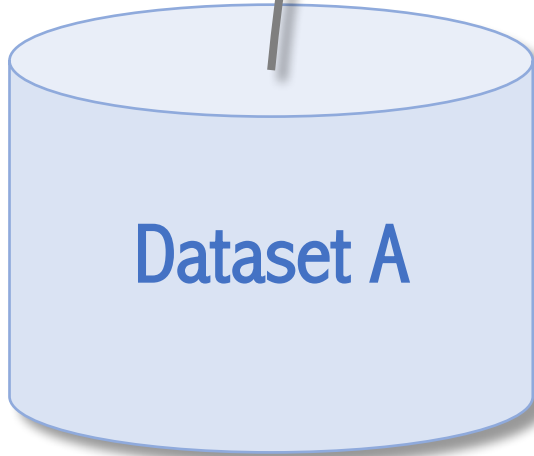
Randomly
Initialized



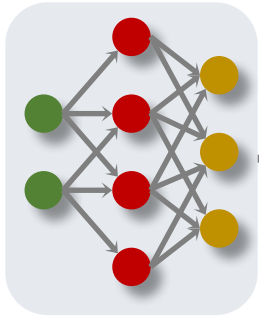
Pretrained



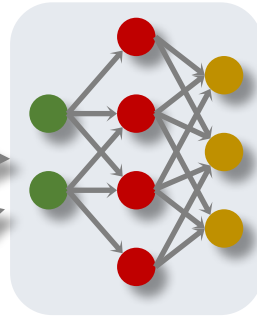
Dataset A



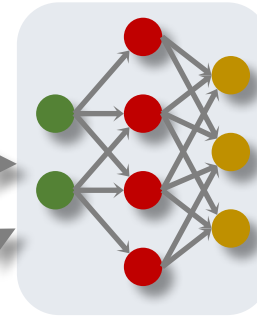
Randomly
Initialized



Pretrained



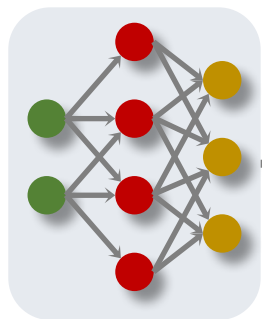
Fine-tuned



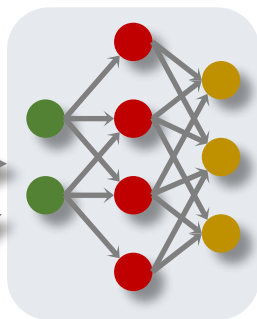
Dataset A

Training Set of
Dataset B

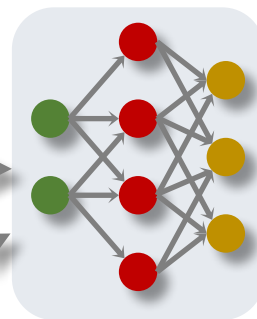
Randomly
Initialized



Pretrained



Fine-tuned

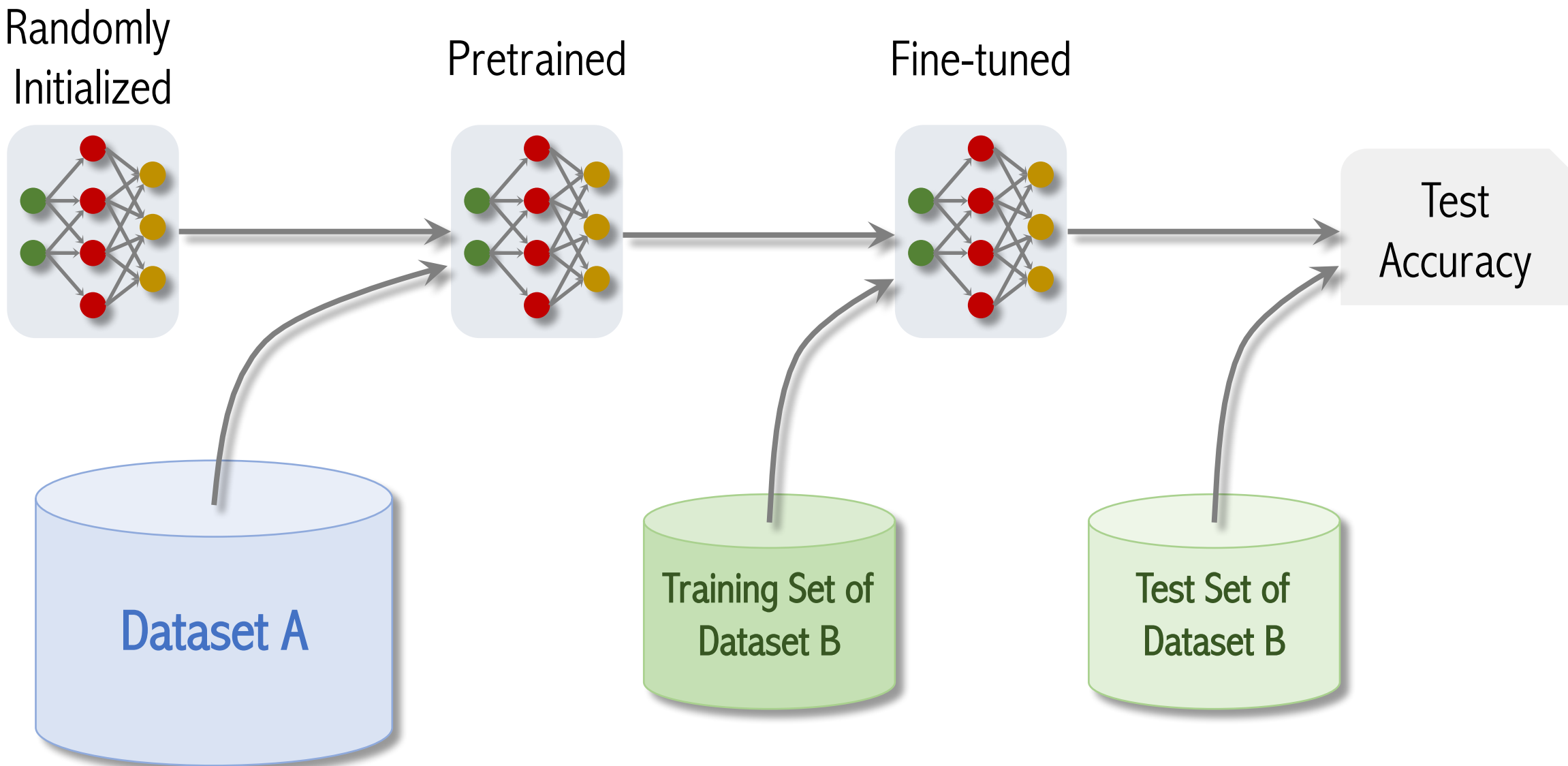


Test
Accuracy

Dataset A

Training Set of
Dataset B

Test Set of
Dataset B



Datasets

	# of Images	# of Classes
ImageNet (Small)	1.3 Million	1 Thousand
ImageNet-21K (Medium)	14 Million	21 Thousand
JFT (Big)	300 Million	18 Thousand

Image Classification Accuracies

- Pretrain the model on Dataset A, fine-tune the model on Dataset B, and evaluate the model on Dataset B.
- Pretrained on ImageNet (small), ViT is slightly worse than ResNet.
- Pretrained on ImageNet-21K (medium), ViT is comparable to ResNet.
- Pretrained on JFT (large), ViT is slightly better than ResNet.

Image Classification Accuracies

ResNet is better

ViT is better

of Images
for
pretraining

100M Images

300M Images



Thank You!